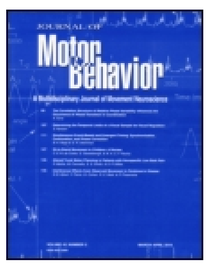
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The Timing of Natural Prehension Movements

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The Timing of Natural Prehension Movements

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ABSTRACT. Prehension movements were studied by film in 7 adult subjects. Transportation of the hand to the target-object location had features very similar to any aiming arm movement, that is, it involved a fast-velocity initial phase and a low-velocity final phase. The peak velocity of the movement was highly correlated with its amplitude, although total movement duration tended to remain invariant when target distance was changed. The low-velocity phase consistently began after about 75% of movement time had elapsed. This ratio was maintained for different movement amplitudes. Formation of the finger grip occurred during hand transportation. Fingers were first stretched and then began to close in anticipation to contact with the object. The onset of the closure phase was highly correlated to the beginning of the low velocity phase of transportation. This pattern for both transportation and finger grip formation was maintained in conditions whether visual feedback from the moving limb was present or not. Implications of these findings for the central programming of multisegmental movements are discussed.

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STUDIES DEALING with visually directed arm movements have usually been limited to simplified actions involving a single joint, or more recently, two joints (Soechting & Lacquaniti, 1981; Abend, Bizzi, & Morasso, 1982). Although these studies are essential for a precise description of the joint torques and the resulting trajectories, they are still far below the complexity of most of our natural actions. Prehension for instance, is a superimposition of two distinct and coordinated movements, that of directing the arm at the spatial location of a target, and that of grasping a three-dimensional object, each movement involving a larger number of joints.

In fact, identification of two distinct components in prehension may have more than a simple descriptive value and bear some relation to the underlying mechanisms. The reaching or *transportation* component involves proximal joints and muscle groups, and reflects determination by the visual system, of the coordinates of a point in a body-centered space. The grasping, or *manipulation*, component involves distal joints and muscles and reflects visual computation of shape, size, and weight of the object. Studies in monkeys have shown that specific brain lesions can alter one of these components without affecting the other (Brinkman & Kuypers, 1972). Developmental studies in human infants

indicate that reaching and grasping do not mature at the same post-natal time and may remain rather independent from each other before they become fully coordinated (Halverson, 1931; Twitchell, 1970; Di Franco, Muir, & Dodwell, 1978). For a complete account of these arguments, see Jeannerod and Biguer (1982).

In the present paper, transportation and manipulation components will be first analyzed separately. A kinematic description based on the profile of tangential velocity can be applied to the transportation component, the pattern of which is closely similar to that classically reported for any accurate aiming movement (Woodworth, 1898; Annett, Golby, & Kay, 1958; Beggs & Howarth, 1972). Only qualitative descriptions, however, are available in the literature for the manipulation component. These descriptions, generally obtained in babies, have established that finger posturing is largely anticipatory and occurs during the transportation itself (see for instance, Trevarthen, 1975). Recently, by using a more precise observation technique, I have found a highly consistent pattern for the respective changes in position of the fingers during the arm movement: fingers first stretch and shape according to the object's shape and size, and then close in anticipation to contact (Jeannerod, 1981). The beginning of the closure phase, which appears to have a fixed time position with respect to movement duration, will be used here as a quantifiable index of manipulation timing. A similar anticipatory closure of the fingers has also been described (though not quantified) in catching movements (Alderson, Sully, & Sully, 1974).

Coordination between the two components is an essential aspect of prehension, particularly if one considers the high degree of precision required when the hand has to grasp small or fragile objects. The problem of coordination between simultaneously moving segments or limbs has been reinvestigated recently (e.g., Kelso, Southard, & Goodman, 1979; Lacquaniti & Soetching, 1982). The main conclusion reached by these studies is that the different segmental components have a common timing. I will show that this is the case also for prehension, but in a limited sense, since synchrony of transportation and manipulation components is reduced to a few points on the time axis. This result will add to the growing evidence for a mechanism maintaining temporal invariance of complex goal-directed actions, as the basis for coordination between moving segments.

Finally, the role of vision in the control of the segmental components and of their coordination, will be studied by varying the degree of visual feedback available from the moving limb. The overall duration of a a prehension movement (when executed without formal time constraint) should normally allow enough time for visual feedback to operate. Indeed the present results will confirm the importance of visual control in improving movement accuracy (Prablanc, Echallier, Komilis, & Jeannerod, 1979), but will show its lack of influence on movement patterning and intersegmental coordination. This difference may be useful in distinguishing between those aspects of an action which are dependent upon visual feedback and those which are not.

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Methods

Apparatus

The apparatus consisted of a box, 68 cm high, 100 cm wide, and 70 cm deep, resting on a table. It was divded horizontally by a semi-reflecting mirror into two equal compartments. Subjects were seated in front of the box with their forehead resting on the front panel. They looked through a window within the upper compartment, and placed their right hand in the lower compartment (Figure 1). Three-dimensional target-objects were placed in the lower compartment along the subjects' sagittal axis. These were small (2 cm to 6 cm) solid objects such as a wooden sphere or cube. Distance from the body could be varied.

Three experimental situations were used

Visual-feedback situation. In this situation subjects could see the lower compartment through the mirror. The hand could not be seen before or during the early part of the movement until it came into the subject's field of view; it then remained continuously visible. The blank zone was about 9 cm long. Since movement velocity is the highest during the early part of the trajectory (see Results) this design resulted in a relatively short blanking period. With a mean velocity of 70 cm/s for instance, the hand would become visible some 125 ms after the onset of the movement.

No-visual-feedback situation. In this situation a mask was inserted below the mirror and the lower compartment was no longer visible. Target-objects were displayed from the top of the upper compartment. Since the mirror was placed half-way between the target display and the table, subjects could see in the mirror a virtual image of the object projecting at the table level (Figure 1). Another object identical to that displayed in the mirror was placed directly on the table in exact coincidence with the virtual image seen in the mirror. Thus the subject reached for the virtual image below the mirror without seeing his hand, and met the second, real, object at the expected location.

No-vision situation. In this case subjects could see the target object before they initiated their movement. When the movement was begun, that is, when the hand was raised from the table, the lights were turned off by means of a microswitch attached to the table. The movement was thus performed in the dark.

Subjects and Procedure

Subjects were 7 young right-handed volunteers (4 males and 3 females). During the experiment, subjects had to place their right hand on a starting block near their body axis, with the forearm in the prone position and the fingers semi-flexed. They were required to perform rapid and accurate movements, to grasp the target-object as precisely as possible and to carry it near the starting block. No formal time con-

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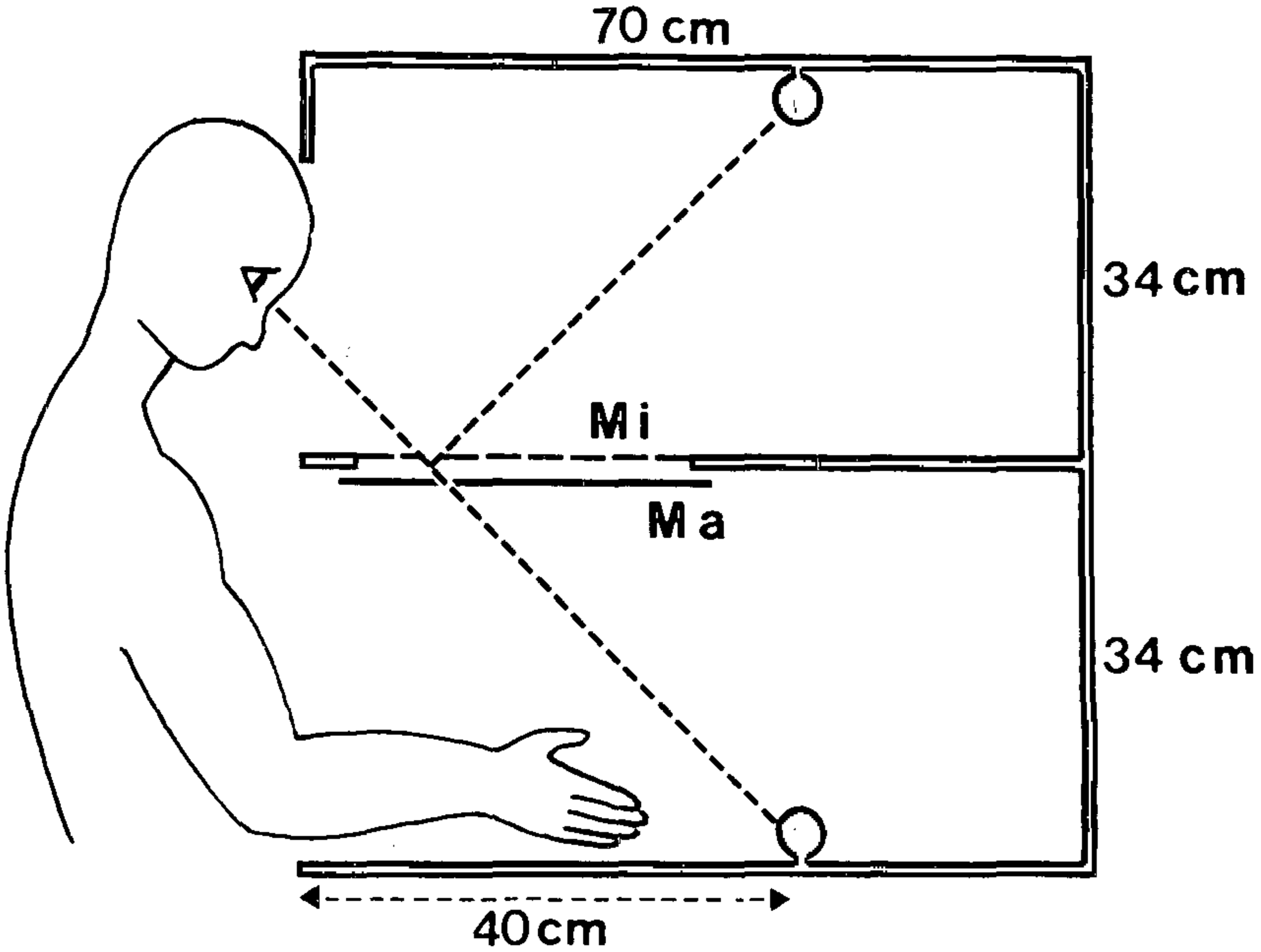


Fig. 1—Experimental set-up for presenting graspable objects.
When the mask (Ma) below the mirror (Mi) is withdrawn the subject sees his hand during part of his movement (visual feedback situation). When the mask is placed below the mirror (no-visual-feedback situation), the hand is no longer visible.

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straint was given. At the beginning of each trial a new object was displayed while the subjects kept their eyes closed. At an acoustic signal they opened their eyes and had to wait 2 to 10 s until a small light in front of them was turned off before performing the reaching movement.

Each subject first received a few practice trials in the visual-feedback situation. The idea was that subjects would be able to stabilize their velocity-accuracy tradeoff before the experimental trials started. Each subject then received a total of 20 experimental trials.

Several experimental conditions were used in different combinations according to subject. The shape of the object was changed across trials for subjects 1–4, but remained constant for the 3 other subjects. Distance of the target was changed across trials for subjects 5–7 (distance set at either 25, 32, or 40 cm from the body); in the other subjects (1–4) the target was maintained at a constant distance (40 cm) from the body. Finally the conditions of visual feedback were changed for 4 subjects: for subjects 1 and 2 the visual-feedback and the no-visual-feedback conditions were used; for subjects 6 and 7, the no-visual-feedback and the no-vision conditions were used. For subjects 3–5 only, the no-visual-feedback condition was used. Pseudorandom rotation of conditions was used with the aim to avoid contamination of the data by sequential effects.

Data from about half of the movements of subjects 1–5 have already been included in a preliminary study (Jeannerod, 1981).

Data Analysis

The radial aspect of the subjects' hand (see Figure 2) was filmed with a cine-camera running at 50 frames/s. Data were processed by projecting the image of the movement frame by frame on a screen, with a 1 × 1 magnification. Duration of the movement was measured as the number of frames between the first detectable arm displacement and contact with the target-object. Data from each second frame were kept for further analysis (i.e., the time resolution was 40 ms).

The position of anatomical details on the wrist was plotted over successive frames. Distance between successive positions gave a measure of the instantaneous tangential velocity for the arm trajectory. From the same frames the relative positions of the tip of the index finger and the tip of the thumb were also plotted. This gave a measure of the size of the finger grip and of its change over time. Due to the resting posture imposed on the hand and the shape of the objects, no rotation of the wrist occurred during the movement. It was assumed that the arm trajectory (and consequently the finger displacement) was rectilinear in the proximo-distal dimension, that is, was effected in a plane parallel to the plane of the film. As far as it could be verified in the present study this appeared to be a reasonable assumption.

Those trials where the fingers happened to come in contact with the table before contacting the object (which could be noticed only during the processing of the film), were discarded. A total of 116 fully analyzed movements from the 7 subjects, (i.e., 16, 14, 19, 14, 14, 19, 20 movements in subjects 1–7, respectively) were retained and are considered in this study. Position, velocity, and acceleration profiles were built for the arm trajectory and the finger-grip size of each movement, with the help of a graphic program on a MINC II computer.

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Results

A. *Reaching movements in the no-visual-feedback situation*

Reaching movements will be first described as they appear in the no-visual-feedback situation, a situation which was available in all subjects. Comparison of movement parameters in different situations will be made in section B.

1. *Duration of the movement.* Duration (T) was found to vary across subjects. For movements directed at targets located at 40 cm, mean duration ranged from about 674 ms for subject 2 up to about 1013 ms for subject 7. This large variability can be explained by the fact that no time constraints were included in the task. However, intrasubject variability was relatively low since coefficients of variation (CV, $S.D./M \times 100$) were within 8% to 13%, except in subject 4 where CV was 17.6% (Table 1). The relation of duration to target distance was tested in

Table 1

Mean duration (*T*, ms) and peak Velocity (*V*, cm/s) and standard deviations of the arm trajectory for 40 cm movements in the no-visual-feedback situation. *n* : Number of reaches. *r*, Pearson correlation coefficient.

Subjects	1 *n* = 7	2 *n* = 7	3 *n* = 19	4 *n* = 14	5 *n* = 6	6 *n* = 6	7 *n* = 6
T (ms); *M*:	708.5	674.2	787.3	960.0	740.7	813.3	1013.3
SD:	64.1	70.9	106.7	169.7	65.7	70.0	123.0
V (cm/s); *M*:	56.0	58.0	75.2	59.2	82.0	69.0	69.0
SD:	4.7	3.1	10.2	5.4	6.0	8.4	5.4
V = *f* (T); *r*:	−0.22 *n s*	−0.80 *p* .01	−0.75 *p* .001	−0.72 *p* .01	0.11 *n s*	0.24 *n s*	0.60 *p* .05

subjects 5, 6, and 7. Coefficients of correlation were 0.24, 0.39, 0.76, respectively (linear regression test). The correlation was found to be weakly significant in subject 7 only ($p < .01$).

2. *Trajectory of the arm.* The general pattern of the transportation component was that of a U-shaped trajectory. The hand was first raised from the resting position and subsequently lowered down to the object. Changing the distance of the target (in subjects 5, 6, 7) apparently did not influence this general pattern. Figure 2 shows a typical example in subject 5.

The profile of tangential velocity (*V*) was asymmetrical. It was consistently marked by a sharp rise up to a peak followed by a less steep

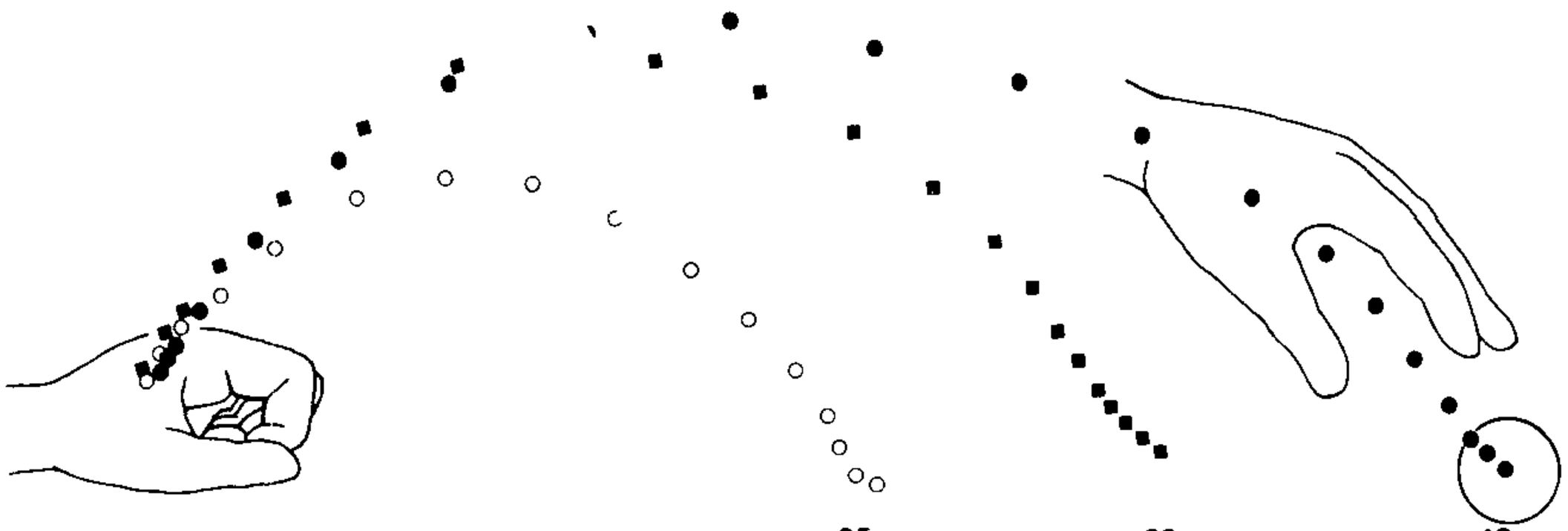


Fig. 2—Pattern of the arm trajectory during the transportation component. Three movements from subject 5 have been plotted. The marks represent the position of the wrist every 40 ms. Movements performed in the no-visual-feedback situation.

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deceleration (see Figures 3 and 5 for sample trajectories). Mean peak velocity ranged between 56 cm/s and 82 cm/s for 40 cm movements (Table I), with peak velocities over 100 cm/s in individual examples. Peak velocity appeared to be poorly correlated to movement duration on the overall sample of movements. However, when this was studied in individual subjects a clear correlation appeared in subjects 2, 3, and 4: (the shorter the movement duration, the higher the peak velocity), although no such correlation was observed in the other 4 subjects (Table 1).

In the 3 subjects where this could be tested, that is in subjects 5, 6, and 7, peak velocity was found to be highly correlated to target distance. Correlation coefficients were 0.91, 0.93, 0.92 respectively ($p < .001$) (Figure 4). It is interesting to note that peak velocity can covary strongly with movement amplitude and weakly with movement duration (e.g., Figure 6 left). This is consistent with the poor correlation between amplitude and duration as already mentioned for the same 3 subjects. However, I do not know whether the same would have been observed for those subjects for whom peak velocity was found to correlate with movement duration.

The deceleration of the arm trajectory was consistently marked by a break point where the tangential velocity tended to become constant or even to re-increase before the movement was stopped at the contact with the object. This break point, which occurred within 70% to 80% of completion of the movement, corresponded to a clear re-acceleration of the arm. Its time value from the origin (*A*) was measured for each movement on the curve of acceleration versus time displayed graphically by the computer. The criterion for the determination of *A* was the value on the time axis of the lowest point on the acceleration graph. Individual examples taken from each of the seven subjects are shown in Figure 5. An unambiguous measure was obtained from 110 movements over the total sample of 116. In 6 cases *A* could not be measured because of the presence of several deceleration peaks of equal value. The retained 110 movements are included in Table 2 which gives in each subject the mean time values of *A* in percent of movement duration ($A/T \times 100$). Notice that movements of different amplitudes and performed in different conditions of visual feedback have been pooled. I felt authorized to do so because of the fact that the populations of A/T values corresponding to different amplitudes and conditions were not different statistically at the 0.1 level of significance (*t* test). Of course, this point will become an important one in discussing temporal invariance of prehension movements. As also indicated in Table 2 a high degree of correlation was found between the time value of A and movement duration.

The time elapsed between A and the end of the movement corresponds to the low-velocity phase of the transportation component. It is clear from the above results that its duration (*T*-*A*) in each individual movement represents a relatively constant ratio of *T*.

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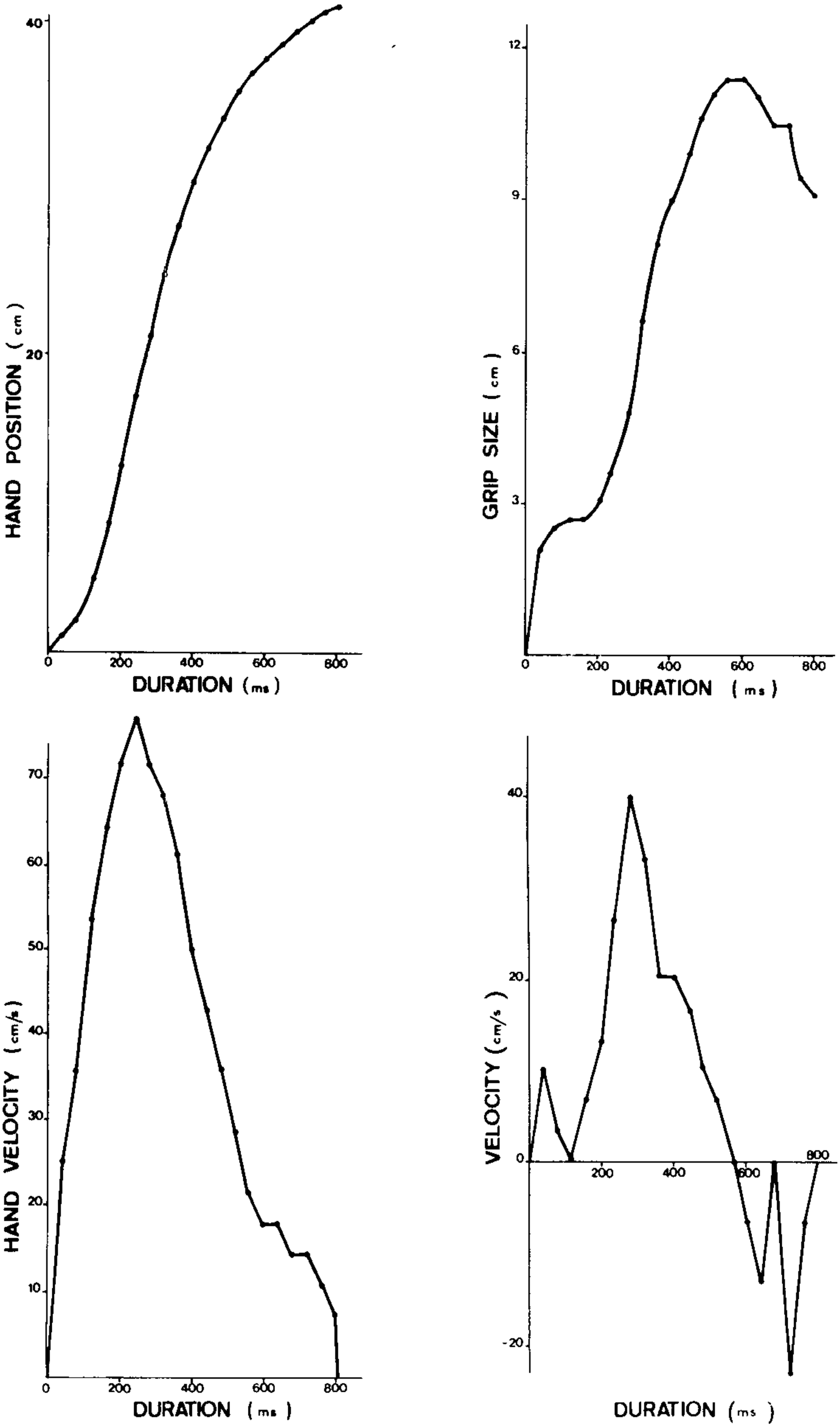


Fig. 3—Position and velocity profiles of the two components of a reaching movement as a function of time.

Left: arm trajectory. The arrow indicates the onset of the low-velocity phase occurring at 560 ms, that is, in this particular case within 70% of completion of the movement (duration, 800 ms).

Right: finger grip. Maximal grip aperture occurs also at 560 ms, that is, within 70% of movement completion.

Movement from subject 1 performed in the no-visual-feedback situation. Distance of the target = 40 cm.

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3. *Finger grip.* The change in finger grip size over time will be also described in this section as it occurs in the no-visual-feedback situation. Since this change takes place during the displacement of the arm, an attempt can be made to relate the temporal aspects of the finger grip to those of the arm trajectory.

The resting position imposed on the subject's hand before performing the movement implied a semiflexion of the fingers. As the arm displacement was started, the fingers began to stretch and the grip size increased rapidly up to a maximum aperture. At a later stage, the fingers were flexed again and the grip size was reduced in order to match the size of the target-object (see Figure 3 right and 6 right). The size of the maximal grip aperture was a function of the anticipated size of the object, i.e., it was larger when the movement was directed at a large object.

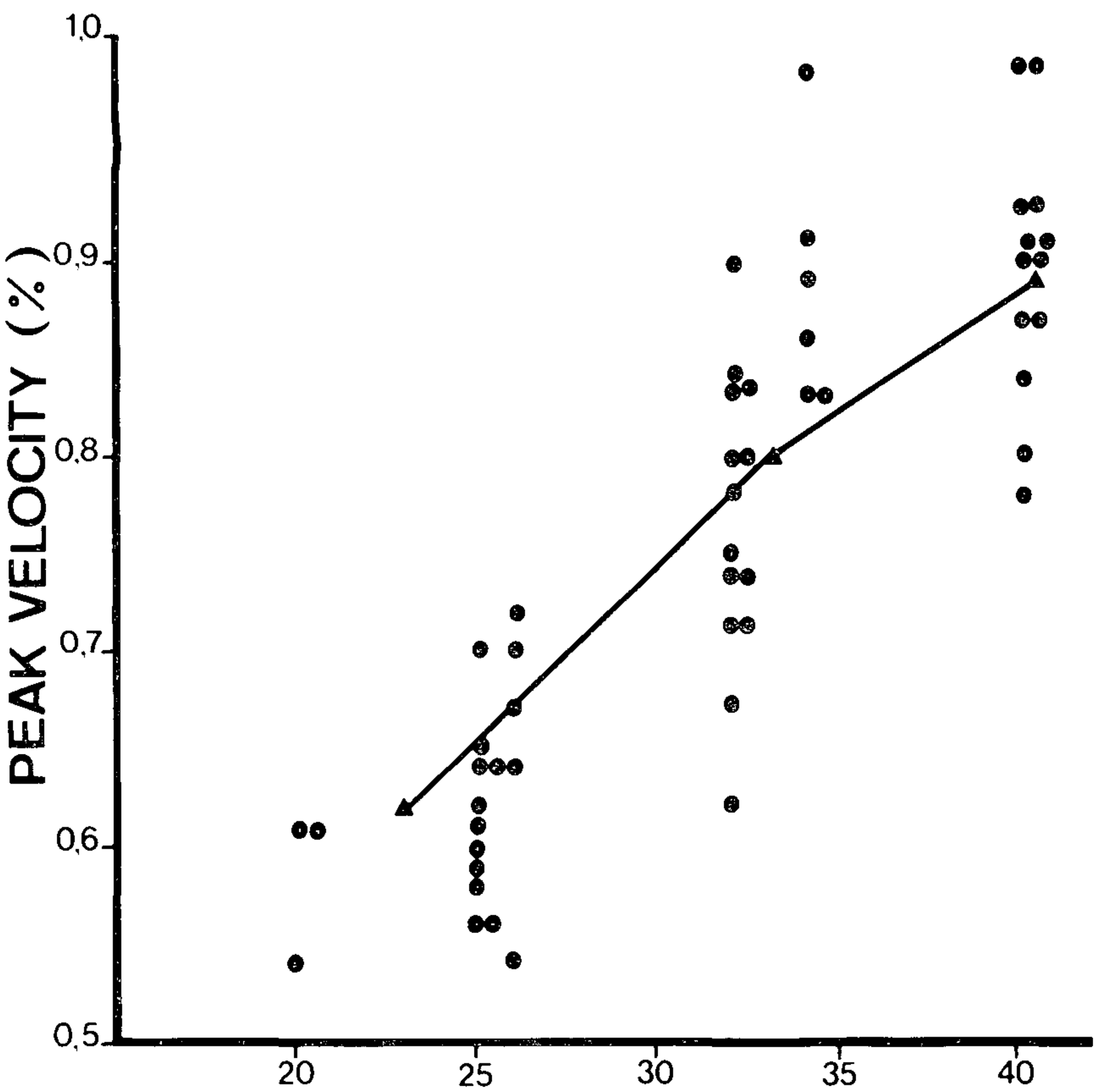


Fig. 4—Relationship of mean peak velocity of the transportation component to distance of the target. Values of peak velocity have been normalized with respect to the highest value for each subject. Solid line, no-visual-feedback situation (subjects 5, 6, 7). Dashed line, no-vision situation (subjects 6, 7).

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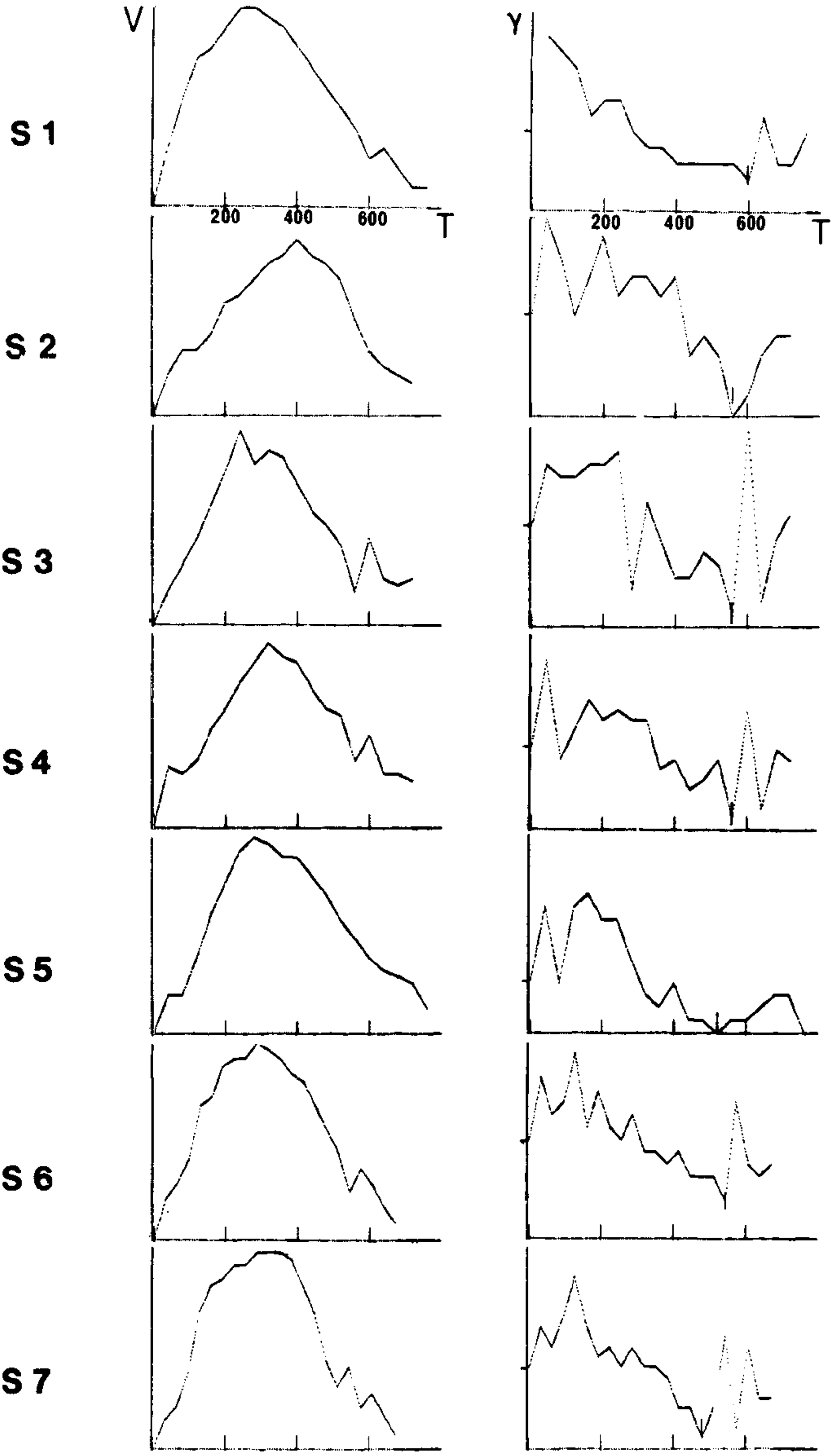


Fig. 5—Computer display of velocity (V, left) and acceleration (γ, right) versus time (T) curves for one reaching movement in each of the seven subjects (S1 . . . S7). For the purpose of illustration, the curves have been smoothed by using a polynomial approximation.

Velocity and acceleration are expressed in arbitrary units.

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The relationship of the grip size to the object size has been described in a previous study. In that paper (Jeannerod, 1981) emphasis was put on the fact that the transportation component remained unaffected when the prehension movement was directed at objects of different sizes.

The average position in time (B) of the point of maximal grip aperture as expressed in percent of T, (B/T × 100) was within 74% to 81% of completion of the movement on the average, except in subject 4 where it occurred earlier (Table 2). In addition a high degree of correlation was found between the value of B and movement duration. This fact appears in Table 2 and is also illustrated by Figure 6 (right), where the profiles of finger grip size over time are shown for 3 movements of different amplitudes (but of the same duration): the B values can be seen to be very similar for the 3 movements.

The time values corresponding to B for each given movement were found to be very close to those corresponding to A for the same movement. As an example A and B have been plotted against each other for the 20 reaching movements of subject 7 (Figure 7): in this case the correlation coefficient for $A = f(B)$ was 0.89 ($p < .001$). The same degree of significance was obtained with five other subjects (r ranging between 0.76 and 0.88). In the last subject (subject 1), r was 0.49 ($p < .1$).

Movements performed in the no-visual-feedback situation often fell short by 1 or 2 cm with respect to the precise location of the target object. In that case, contact of the hand with the object occurred at the level of the very tip of the forefinger and final prehension was achieved

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Table 2

Relationship of parameters *A* (onset of the low velocity phase of the arm trajectory) and *B* (maximal finger grip aperture) to movement duration (*T*). *n* : number of reaches. *M*: mean time occurrence of *A* or *B* in percent of *T*. *SD* : standard deviation; *r*, Pearson correlation coefficient. In this table movements of different amplitudes and performance in different feedback conditions have been pooled.

Subjects	1 *n* = 14	2 *n* = 14	3 *n* = 17	4 *n* = 12	5 *n* = 14	6 *n* = 19	7 *n* = 20
A/ *T*x100; *M*:	73.8 %	71.9 %	75.8 %	66.2 %	74.5 %	77.7 %	71.3 %
SD:	11.8	11.3	11.8	17.4	6.4	6.4	9.9
A = *f*(T); *r*:	0.20	0.68	0.74	0.78	0.71	0.86	0.83
	ns	*p* .01	*p* .001	*p* .001	*p* .01	*p* .001	*p* .001
B/*T*x100; *M*:	75.9 %	74.9 %	75.1 %	61.4 %	81.7 %	81.4 %	79.6 %
SD:	10.3	6.0	8.5	9.3	6.1	5.3	9.9
B = *f*(T); *r*:	0.49	0.81	0.71	0.84	0.76	0.88	0.89
	p .1	*p* .001	*p* .001	*p* .001	*p* .001	*p* .001	*p* .001

Fig. 6—An illustration of invariance of the time of occurrence of maximal finger grip in movements of different amplitude and same duration.
Left: velocity profile of the transportation components.
Right: position profile of the finger grips.
Movements from subject 6 performed in the no-visual-feedback situation.

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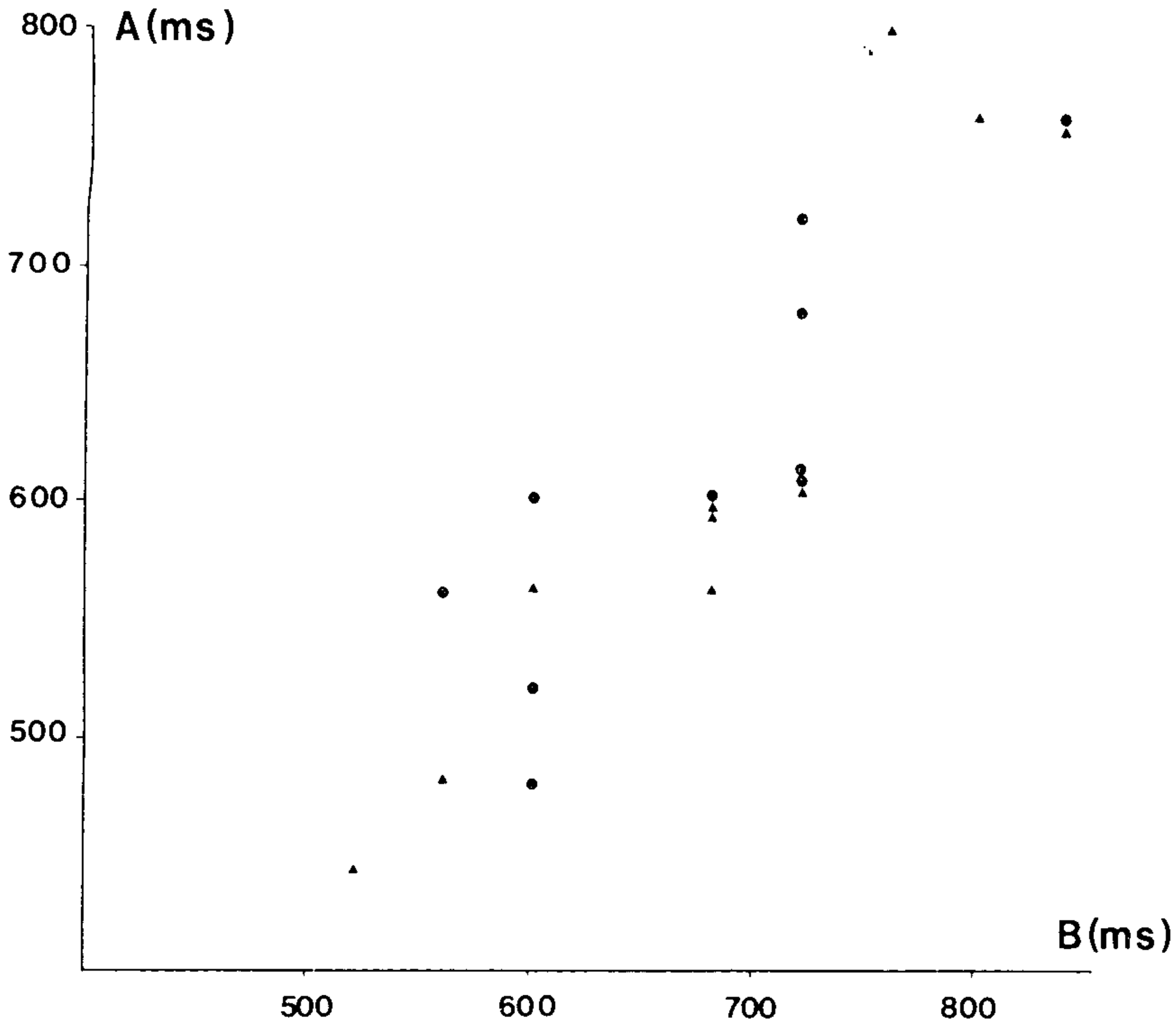


Fig. 7—Correlation between the onset of the low velocity phase of the transportation component (parameter *A*) and the time occurrence of the maximal grip aperture (parameter *B*) in subject 7. Note similar distribution for movements performed in the no-visual-feedback situation (circles) and in the no-vision situation (triangles).

through a brief reshaping of finger posture under tactile guidance. In the normal situation (i.e., in the presence of visual-feedback) contact with the object occurs simultaneously at the level of the palmar surface of the thumb and of the forefinger, at the end of finger closure. Since undershooting the target is a common finding for movements of a large amplitude performed without visual feedback (see Prablanc et al., 1979), I have not analyzed further this aspect of the data.

B. *Influence of visual control on reaching movements*

The above description dealt exclusively with prehension performed in the no-visual feedback situation, that is, the subject could not see his hand. In this section, these results are compared to those obtained in two other situations: one where visual feedback from the moving limb is present and one where no visual cue is present.

1. *Reaching movements in the visual-feedback situation.* This situation was used in subjects 1 and 2. Thus a comparison can be made with the results obtained in the no-visual-feedback situation in the same subjects.

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In Table 3 are reported the mean values of movement duration and peak velocity of the arm trajectory in the two situations. It can be seen that movements performed with a visual control of the trajectory had a significantly longer duration. Peak velocity was found to be lower for subject 1, not for subject 2 (Table 3). The mean time value *A* of the velocity break point corresponded to 72.3% (*SD*, 8.6) of movement duration for subject 1, and 66.4% (*SD*, 5.6) for subject 2. In the no-visual-feedback situation those values were 75.4% (*SD*, 9.2) and 77.5% (*SD*, 6.3) respectively. The slight decrease in the value of *A*, together with the increase in movement duration in the visual feedback situation resulted in an increase of the average value of the low-velocity phase (*T-A*) in this situation. In subject 1, *T-A* was about 191 ms in the no-visual-feedback situation and about 225 ms in the visual feedback situation (148 ms and 240 ms for subject 2).

The qualitative aspects of finger grip were not modified by visual control. In the visual feedback situation the mean time value *B* of the maximum grip size correspond to 75.9% (*SD*, 8.4) of movement duration for subject 1 and 73.6% (*SD*, 4.2) for subject 2. These values are to be compared to those obtained in the no-visual-feedback situation, that is, 75.9% (*SD*, 7.8) for subject 1 and 76.3% (*SD*, 4.7) for subject 2. In all cases the duration of the finger closure phase thus remained closely correlated to that of the low-velocity phase of the transportation component.

2. *Reaching movements in the no-vision situation.* This sitatuion was used for subjects 6 and 7. When the results were compared with those obtained in the no-visual-feedback situation for the same subjects no difference could be found. This allowed pooling the data from the two situations in Tables 1 and 2. In fact, when data from the no-vision situa-

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Table 3

Comparison of mean duration (*T*, ms) and peak velocity (*V*, cm/s) of the arm trajectory in the visual-feedback and in the no-visual-feedback situations, in subjects 1 and 2. 40 cm movements. *n* : number or reaches.

Subjects	1		2	
Experimental Situation	Visual feedback $n=9$	No-visual feedback $n=7$	Visual feedback $n=7$	No-visual feedback $n=7$
T (ms); *M*:	782.3	708.5	714.2	674.2
SD:	60.3	64.1	74.5	70.9
V (cm/s); *M*:	48.2	56.0	59.2	58.0
SD:	3.3	4.7	4.7	3.1

tion were analyzed separately, the correlation of movement duration to target distance was found to be non-significant for subject 6 and slightly significant for subject 7. The correlation between peak velocity and target-distance was present in both cases ($r = 0.77$ for subject 6, $r = 0.80$ for subject 7) (see Figure 4).

Similarly, the mean time values of parameters *A* and *B* were within the same range as in the no-visual-feedback situation for both subjects. Figure 7 displays the values of $A = f(B)$ for movements performed in both the no-visual-feedback situation and the no-vision situation for subject 7. The two distributions are clearly superimposed.

C. *Preliminary observations on bimanual movements*

A short series of prehension movements performed with the two arms was studied in subjects 6 and 7. Though still incomplete, these observations are worth reporting because they provide useful information on intersegmental coordination during prehension.

The study was carried out with subject 6 and 7 during independent experimental sessions. Subjects had to reach for a small bottle placed at 40 cm in front of them, and remove the plug placed at its top. This action required grasping the body of the bottle (6.5 cm diameter) with one hand and removing the plug (0.6 cm) with the other hand. Only the no-visual-feedback situation was used.

At the beginning of a trial, the subject had to place his two hands in the already described resting position on each side of his body. Other aspects of the procedure were as in the unimanual experiment. Two cameras were used: one was placed to the side of the box as already described, and was used to film the radial aspect of the right hand; the other one was placed below the table and was used to film the cubital aspect of the left hand through a transparent shield. The two films were synchronized and processed frame by frame.

From the 7 bimanual movements which could be fully analyzed, it was found that the two arms made remarkably synchronous trajectories. In all cases, they began to move (i.e., the two hands were raised) at the same time, and movement duration was the same. In 5 out of the 7 movements the peak velocity had the same position in time for the two trajectories (see for instance Figure 8 left). In the other 2 cases the velocity of the right and left arms peaked within 80 ms.

In each bimanual movement, the amplitude of the peak velocity was found to be strikingly similar for the two arm trajectories (see Figure 8 for a typical example). This result is another illustration of the relationship of peak velocity to target distance as already observed for unimanual movements (see section A, Table 1, Figure 4).

The grip size and shape were found to be very different for both hands due to the difference in size and orientation of their respective target-object. Nevertheless, the points of maximum grip aperture (B values) appeared to be synchronous for the two hands in 3 cases and were within 80 ms or less in the 4 others. In addition, the correlation between

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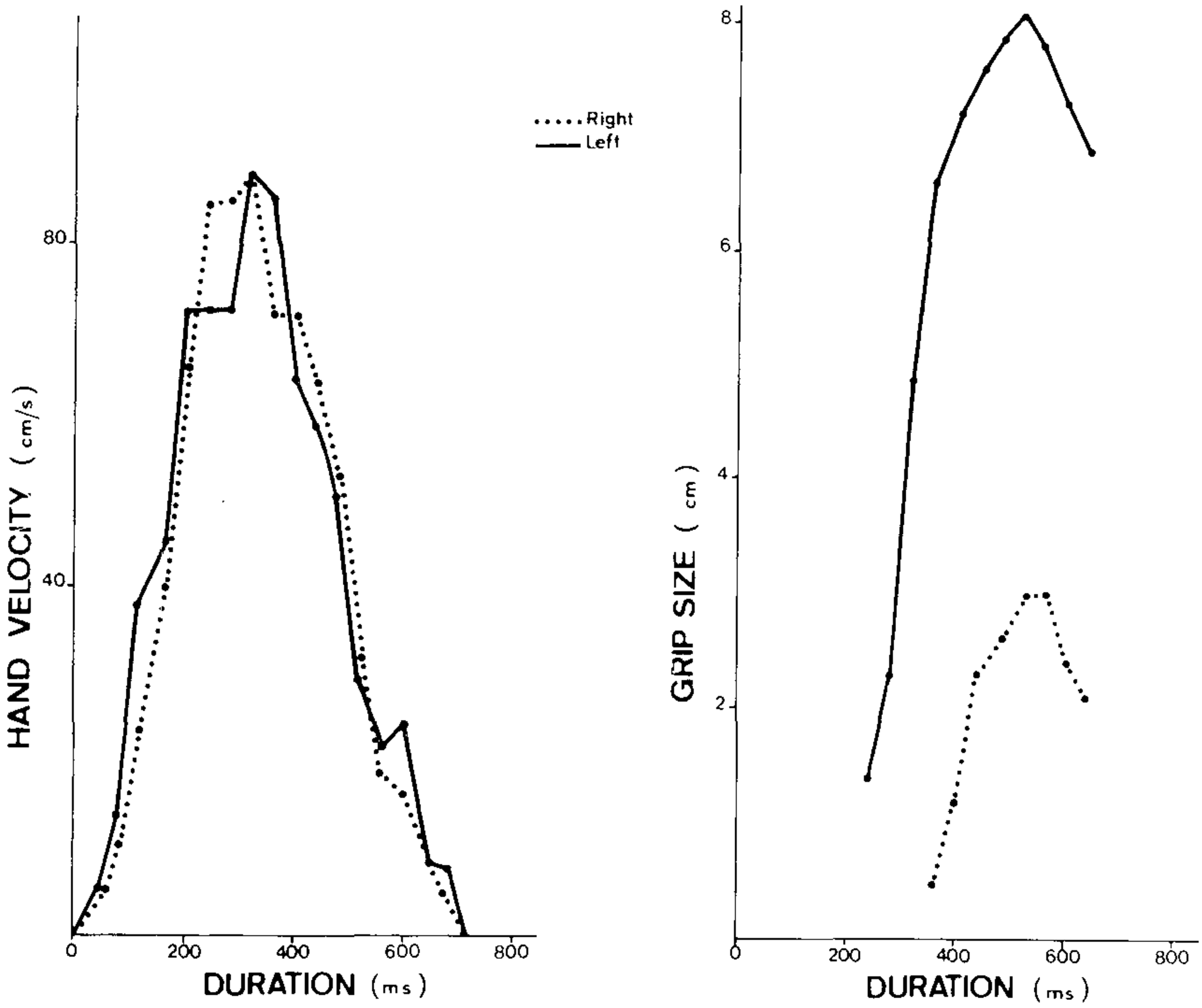


Fig. 8—Pattern of intersegmental coordination during a bimanual prehension movement.

Left: velocity profiles of the transportation components of the right (dotted line) and of the left arms (solid line).

Right: position profiles of the two corresponding finger grips.

The initial part of the finger grips could not be plotted due to shadowing on the film. Movement performed in the no-visual-feedback situation. Subject 7.

these values of B and the values of A (onset of the low-velocity phase of the arm trajectory) was consistent with that obtained in unimanual movements (Figure 8).

Discussion

The main question raised by the present rules is how the transportation and the manipulation components are coordinated in time, in order to achieve a single prehension movement. My hypothesis will be that the timing of the two components is achieved by a centrally generated temporal pattern and that this pattern remains uninfluenced by feedback control.

1. *Invariant duration of the transportation component*

It is a common finding that velocity of a movement increases with its amplitude. This fact, first reported for saccadic eye movements (Becker

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& Fuchs, 1969) has been also clearly observed for arm movements. Such behavior could be a way of achieving invariant movement duration, since increasing the speed of a moving segment within a fixed duration should automatically yield a larger displacement. In fact, in the case of eye movements, as in the case of simple, relatively fast, arm movements like pointing or hitting a target, not only the velocity but also the duration of the movement is positively correlated with the target distance (Fitts, 1954; Beggs & Howarth, 1972; Prablanc et al., 1979; Wadman, Denier van der Gon, Geuze, & Mol, 1979). Invariant duration seems to have been observed only in the case of more complex movements executed without enforced time constraints. Viviani and Terzuolo (1980), for instance, have shown that writing the same letter at different sizes is achieved by simply changing the tangential velocity of the writing movement without changing its duration (see also Michel, 1971; Viviani & Mc Collum, 1983). Prehension movements such as those studied here would appear to belong to this category. Indeed peak velocity was found to correlate strongly with target distance, although duration was found to remain more or less constant, at least within the range of amplitudes that were tested.

Another feature of the prehension movements relevant to this point is the relatively invariant ratio of movement time devoted to the low-velocity phase. For all subjects but one the mean values of A/T were clustered within about 75% $\pm$ 3% of movement duration. Also, changing the target distance (for 3 subjects) did not influence the value of A/T, that is, the duration of the low-velocity phase remained a fixed proportion of movement duration.

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2. *Invariance of the relative timing of the grasp and transportation components*

There is in the central patterning of movement a general trend toward synchronization of the neural commands related to a single action (see for instance Biguer, Jeannerod, & Prablanc, 1982). This type of organization, in phasing the segmental movements, must necessarily result in some kind of temporal invariance. Such an arrangement at the programming level may have several interesting advantages. First, it may reduce the complexity of the system by bringing the number of time variables to be controlled down to a minimum (Soechting & Lacquaniti, 1981). It may also ensure a faster mobilization of the motor ensemble directed at the same goal. In the case of prehension, two segmental movements which differ widely in their velocity profile over time are superimposed. In order to achieve coordination between these two movements it appears necessary first, to synchronize their onset, and then to maintain a correspondence between the two trajectories. This is indeed what is observed in prehension movements where synchronization seems to be obtained by a double clustering of the motor commands. The first burst of commands synchronizes the fast phase of the transportation component with extension of the fingers; the second burst synchronizes the

low-velocity phase (which in fact corresponds to a secondary acceleration of the arm movement) with finger closure. Similar examples can be found in Kelso et al. (1979) for interlimb coordination and in Lacquaniti and Soechting (1982) for coordination of two joints of the same arm during a pointing task.

One major interest in this arrangement at the programming level is to keep the timing of the commands independent from the spatial parameters of the movement (Bernstein, 1967; Arbib, 1980). In other words, the selection of muscles to be activated can be modified or the torque applied to the joints can be modulated, without affecting the temporal template which determines the coordination for a given action. The observation made by Fraser and Wing (1982) in a subject using one artificial arm seems to be relevant to this point. In studying the reaching and grasping movements made by the subject with that arm they had observed that the time relation between the arm trajectory and grip formation was preserved to about the same extent as with the other, normal, hand. This coordination however was effected in a strikingly different way for each hand. The subject operated his artificial hand by a shoulder harness and hand opening was obtained by shoulder protraction. Hence, not only were the spatial parameters of the movement completely different from those of the normal hand; also the selection of agonist-antagonist muscle pairs prescribed by the functioning of the artificial hand seems to have precluded a possible sequential (or reflex) activation of the pairs of the different segments. This observation thus seems to validate the hypothesis made at the beginning of this discussion of a centrally generated temporal template as the basis for intersegmental coordination.

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3. *The nature of the low-velocity phase of the transportation component*

The commonly observed low-velocity phase at the end of aiming movements has been attributed by most authors to a "corrective" function. According to this hypothesis, the hand enters the central part of the visual field, during this part of the trajectory. Its final position can be anticipated from its instantaneous velocity and compared to the position of the target. Finally a correction can be generated on the basis of this information.

This mechanism has been demonstrated for saccadic eye movements: the main saccade appears to systematically undershoot target position, and is completed by a second smaller, corrective saccade. The corrective saccade is no longer present when there is no information available on target position at the end of the main saccade (Prablanc & Jeannerod, 1975). The same argument does not hold for prehension movements, since the low-velocity phase is still observed in the absence of visual-feedback, and even in the no-vision situation. This finding, however, does not preclude that visual feedback, when present, will be incorporated during the late part of the movement (or at any time during the trajectory) in order to smooth precision. Movements performed

without visual feedback lack accuracy (Prablanc et al. 1979). In the present study, movement duration and low-velocity phase duration were found to be increased in the visual feedback situation. This additional time might represent the time needed to process visual information gained during the movement.

Another alternative for the role of the low-velocity phase is that it would still represent a corrective movement, but be triggered by proprioceptive feedback. This suggestion, however, does not fit the present data. The low-velocity phase remains an invariant proportion of movement duration for different movement amplitudes. Since the amount of proprioceptive feedback generated by these movements must be a function of peak velocity and joint angles, it is hard to imagine how the low velocity phase could be triggered by such a variable signal at a rather constant point in time.

My final suggestion is that the low-velocity phase is not a corrective movement based on feedback processing at the input level, but pertains to the central patterning of prehension. As already suggested by others it would represent a positioning (Annett et al., 1958), or target-acquisition phase (Welford, Norris, & Shock, 1969), a necessary constraint for movements involving several segments and requiring a high degree of precision (like grasping small or fragile objects). In the present results, the synchronization of the low-velocity phase with finger closure indicates that both are controlled by a common program which achieves the timing of coordination.

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NOTE

1. In analyzing the data these hypometric movements have been considered as belonging to the same class of amplitude (i.e., 25, 32, or 45 cm) as those which achieved a precise reach.

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